Dhruv Sharma

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Research Interests

Numeric transformers for continuous signals; LLM agents that write and verify code; reinforcement learning and sim-to-real transfer for robots and drones. Long-term goal: a "nervous system" of models for a robot, with numeric models at each joint predicting the rotation and torque it needs, vision models as its eyes, and a supervising "brain" model that makes the final decisions.

Education

MITx MicroMasters Program in Finance, Massachusetts Institute of Technology (online) — Oct 2025 – Feb 2027 (expected)
Completed: Mathematical Methods for Quantitative Finance (Jan 2026); Derivatives Markets (May 2026)

Post Graduate Program in Algorithmic Trading, Indian Institute of Quantitative Finance — Oct 2025 – Jun 2026

Bachelor of Technology, Computer Science & Engineering, Arya Institute of Engineering, Technology & Management, Jaipur (affiliated with Rajasthan Technical University, Kota) — 2020 – 2024
First Division with Distinction; final-year project graded A++. English-medium instruction. Worked full time in industry alongside the degree from 2022.

Senior School Certificate (Science), Malviya Convent School, Jaipur · CBSE — 2018 – 2020
Cleared the Pre-Regional Mathematical Olympiad (PRMO) 2018 and qualified for the Regional Mathematical Olympiad (RMO), the only student from the school to do so.

Research

FDT-N: Footprint Distribution Transformer — D. Sharma · paper · code — Manuscript, 2026

- 372K-parameter ragged-set transformer over full order-flow footprint rows; three seeds, model selection on validation loss only.
- Held-out q50 MAE 7.25% better than persistence but 2.68% worse than a train-only ridge baseline (ridge won 55/56 horizon-target cells); reported as a negative result. arXiv submission pending endorsement.

CHMT-Q: Causal Hierarchical Market Transformer with Quantiles — D. Sharma · paper · code — Manuscript, 2026

- 2.1M-parameter transformer with separate bar / L1 / L2 encoders, quality-aware fusion and monotone q10/q50/q90 heads; content-hashed data, split-bound scalers, walk-forward with purge/embargo.
- Weighted pinball loss +2.26% over an empirical baseline; directional and cost tests failed, so the model was documented as no-deploy. arXiv submission pending endorsement.

HTT: Hyperion Terminal Transformer — D. Sharma · paper — Architecture paper, 2026

- Design of a 1.52B-parameter decoder-only model for strategy and visualizer code (RMSNorm, RoPE, SwiGLU, GQA, hybrid global/sliding-window attention, grammar-constrained decoding), evaluated by sealed functional correctness with an ablation and stop condition for every design choice.

Selected Projects

DaxAlgo — Founder · team of 3 · daxalgo.ai May 2026 – present

- Built the entire AI-native trading platform alone: a .NET 9 desktop terminal (Linux/macOS port in progress), broker integrations (Interactive Brokers, cTrader, Alpaca), website and marketplace, and cloud-service infrastructure.
- *Hyperion* AI harness: natural-language specifications become strategy code gated by compile → tests → capability checks; a 20-task benchmark scores outputs against commercial reference products.
- DAXQ: a custom compiler and native C++ VM that run users' strategy code without exposing its source; capability-scoped sandbox with a public SDK (v0.4) and a custom Roslyn analyzer.
- Execution engine (OMS, risk engine) over IBKR, NinjaTrader, cTrader and Alpaca behind one broker interface, with two independent live-trading safety gates; strategies can only trade their own virtual book.
- Research tooling: tick-level backtester shared with live trading, C++20 engine for 50M+ tick workloads, CLI for sweeps, walk-forward, Monte Carlo bootstrap and TCA.

Hyperion: Evidence-Gated LLM Coordinator · code 2026

- Five-role agent workflow (Planner, Evidence Analyst, Critic, Synthesizer, Risk Judge) with hash-chained state and human approval; distillation/QLoRA training harnesses, contamination checks, sealed evaluation and an isolated C# compile-and-replay verifier. 553 .NET + 494 Python tests.
- Designed a custom transformer architecture (inspired by Sakana AI's Fugu) and trained a model with it by distilling an open-weight model; improved version in progress, code release planned.

drone-vs-gunner: Adversarial Self-Play from Pixels · code 2026

- Camera-only quadcopter evader vs. turret gunner, trained in simulation for transfer to a real drone and a 720° robotic arm; independent PPO with snapshot-pool self-play, PFSP, ELO, 5-stage curriculum and an asymmetric actor-critic. Trained policies rendered in desert, forest, snow and urban scenes with a PyVista renderer. Full GPU training run in progress.

Max Delta Systems — real-time airspace awareness, defense startup, Germany Sep 2026 – present

- Helping a friend's startup with vision models that detect objects in the sky and estimate their properties in real time, and with a physics engine that simulates them for testing; multi-camera “pixel to voxel” ray intersection to locate and track objects in 3D, fused with live ADS-B data.

Outfit Matcher: From-Scratch ViT for Photo-to-Render Retrieval 2026

- ViT-S/16 trained from scratch (21.9M parameters) with SupCon + auxiliary CE, bf16 DDP on 2×RTX 5090, and domain randomization to close the render-to-photo gap; an earlier ResNet-18 version reached Recall@10 83.5% on DeepFashion In-Shop.

Market Microstructure Strategies — SigmaIcFlow, IndexRegimeGraph and 7 others · code 2026

- Order-flow signals fused with Ledoit–Wolf shrunk covariance, Σ^{-1}·IC weights, isotonic calibration and quarter-Kelly sizing; regime consensus over a stack of market-regime models.

Jane Street Puzzles — solved all five monthly puzzles, Jun–Oct 2026 (named on leaderboards) 2026

Other: RL from scratch (PPO/SAC/DQN in NumPy) · deep learning for price prediction · forex ML research · Tick-BackTester (C++20, 3–6M ticks/s) · QuantChess (perft-verified) · Vision Trainer

Professional Experience

Rockstar Games — C# Full Stack Developer | Associate Technical Artist, Bengaluru Oct 2023 – present

- Production tools for the RAGE engine used across studios; joined full time during the final year of the degree.
- Credited on *GTA V*, *GTA Online*, *Red Dead Redemption 2* and *Red Dead Online*; currently working on *Grand Theft Auto VI* (credited).
- Further details of this work are covered by a non-disclosure agreement.

Kyrel (gaming startup) — C++ Tools Programmer | Unreal Developer, Noida Apr 2023 – Sep 2023

- Built a complete third-person shooter alone in UE5/C++: 15+ state movement machine, hitbox combat, weapons, 8 abilities.
- Multiplayer on EOS and PlayFab (matchmaking, lobbies, latency compensation); 60 FPS with 100+ networked entities via octree, client prediction and server reconciliation; delta compression cut bandwidth 40%.

Lakshya Digital — Python Programmer | Technical Artist, Gurugram Aug 2022 – Feb 2023

- Full-time offer during college. Automated the FIFA MatchDay asset pipeline (1,000+ player cards a day; batch time 8 h → 20 min).
- Three studio-wide Blender plugins (topology QA, input remapper, GPU UV packing 5 min → <30 s); Blender/3ds Max/Perforce batch framework.

Technical Skills

ML / Research	PyTorch (CUDA, DDP), transformers, contrastive learning, RL (PPO, SAC, DQN), distillation/QLoRA, walk-forward and purge/embargo validation, ONNX, Isaac Lab, PyVista
Languages	Python, C#/.NET 9, C++20, TypeScript, SQL
Systems	Compilers and VMs, Roslyn analyzers, sandboxed runtimes, Avalonia/WPF, FastAPI, ASP.NET Core, Cloudflare Workers, Docker

Volunteering

Volunteer, Ota Tokyo event, Bengaluru Jan 2026

- Guided Japanese guests and visitors around the event using self-taught Japanese.

Languages

English (fluent; English-medium education from school through university, and English as the working language at Rockstar Games and DaxAlgo) · Japanese (self-taught; JLPT N5 exam, December 2026)